



$$\underline{7} \quad (p \rightarrow r) \wedge (q \rightarrow r) \equiv (p \vee q) \rightarrow r \quad \text{IMPORTANT-NOTES}$$

$$\underline{8} \quad (p \rightarrow q) \vee (p \rightarrow r) \equiv p \rightarrow (q \vee r).$$

$$\underline{9} \quad (p \rightarrow r) \vee (q \rightarrow r) \equiv (p \wedge q) \rightarrow r.$$

LOGICAL Equivalence involving Biconditionals:

$$\underline{1} \quad p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p)$$

$$\underline{2} \quad p \leftrightarrow q \equiv \neg p \leftrightarrow \neg q.$$

$$\underline{3} \quad p \leftrightarrow q = (p \wedge q) \vee (\neg p \wedge \neg q).$$

$$\underline{4} \quad \neg(p \leftrightarrow q) \equiv p \leftrightarrow \neg q.$$

PREDICATE

→ It is a sentence that contains finite number of variables.

→ It becomes a proposition when specific values are substituted for the variables.

Ex Ram is a Bachelor.

Sohan is a Bachelor.

$p(x)$: x is a bachelor.

x → predicate variable.

$p(x)$ → propositional function.



Ex ~~Let~~ $p(x) : x > 3$, what are the truth values of $p(4)$ & $p(2)$.

$p(4) : 4 > 3$, which is true.

$p(2) : 2 > 3$, which is false.

Ex Let $Q(x, y) : x = y + 3$, what are the truth values of the propositions $Q(1, 2)$ & $Q(3, 0)$.

Sol $Q(1, 2) = 1 = 2 + 3$; which is false.

$Q(3, 0) = 3 = 0 + 3$, which is true.

Ex $p(x) : x^2$ is a positive number for every real no. x .

Sol $p(x)$ is true for every x .

Quantifiers.

Quantifiers are words that refer to quantities such as some, few, many, all, none and indicate how frequently a certain statement is true.

Types of Quantifiers

Universal Quantifiers

1) The phrase "for all" is called Universal quantifiers.

2) It is denoted by $\forall$.

3) $\forall x$ represents "for all x ",
"for every x ",
"for each x ".

Existential Quantifiers

1) The phrase "that exists" is called Existential Quantifier.

2) It is denoted by $\exists$.

3) $\exists x$ represents
"there exists an x ",
"for some x ".

IMPORTANT-NOTES

4 All human beings are mortal.

$p(x)$: x is mortal.

i) can be written as.

$(\forall x \in U) p(x)$ or $\forall x p(x)$

U - domain or universe of discourse

→ All students are successful.

$p(x)$: x is successful.

$\forall x p(x)$

← ~~There is at least one x .~~

Some human beings are mortal.

$p(x)$: x is mortal

$\exists x p(x)$

Some students are successful

$p(x)$: x is successful

$\exists x p(x)$

Translate sentences into logical expressions.

1 Every person is precious.

Rephrase: For every x , if x is a person then x is precious.

$M(x)$: x is a person, $A(x)$: x is precious.

$\therefore \forall x (M(x) \rightarrow A(x))$.

2 Every student is clever.

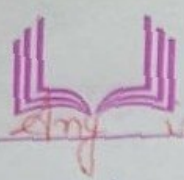
$M(x)$: x is student; $N(x)$: x is clever

$\forall x (M(x) \rightarrow N(x))$.

3 All men are mortal.

$M(x)$: x is a man, $p(x)$: x is mortal.

$\forall x (M(x) \rightarrow p(x))$.



IMPORTANT-NOTE

4 any integer is either +ve or -ve

Q1 $M(x)$: x is an integer $P(x)$: x is either positive or -ve.
 $\Rightarrow \forall x (M(x) \rightarrow P(x))$

5 There exists a student.

$P(x)$: x is a student $\exists x (P(x))$

6 Some students are clever.

$P(x)$: x is a student $Q(x)$: x is clever.

$$\exists x (P(x) \wedge Q(x))$$

7 Some students are not successful.

$P(x)$: x is a student $Q(x)$: x is successful.

$$\exists x (P(x) \wedge \neg Q(x))$$

8 Not all birds can fly.

$P(x)$: x is a bird $Q(x)$: x can fly.

$$\neg (\forall x (P(x) \rightarrow Q(x)))$$

9 There is a student who likes Mathematics but not Geography.

$S(x)$: x is a student.

$M(x)$: x likes Mathematics

$G(x)$: x like Geography

$$\exists x (S(x) \wedge M(x) \wedge \neg G(x))$$

Logical Equivalence involving Predicates and Quantifiers **IMPORTANT-NOTES**

Two logical statements involving predicates and quantifiers are said to be equivalent if and only if they have the same truth value in all possible cases.

Logical equivalence is helpful in replacing one expression with an equivalent expression.

There are 2 important equivalences involving quantifiers

$$\rightarrow \forall x (P(x) \wedge Q(x)) \equiv \forall x P(x) \wedge \forall x Q(x).$$

$$\rightarrow \exists x (P(x) \vee Q(x)) \equiv \exists x P(x) \vee \exists x Q(x).$$

proof

let us consider an example.

Domain: All students of xyz university.

$P(x)$: x has studied discrete mathematics.

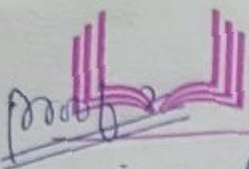
$Q(x)$: x has scored more than 60% marks in exam.

$\forall x (P(x) \wedge Q(x))$: Every student in the xyz university has studied discrete mathematics and has scored more than 60% marks in exam.

$\forall x P(x) \wedge \forall x Q(x)$: Every student in the xyz university has studied discrete mathematics and every student in the xyz university has scored more than 60% marks in exam.

	Discrete math	>60%
Mark	✓	✓
Joe	✓	✓

$\forall x (P(x) \wedge Q(x))$
 $\forall x P(x) \wedge \forall x Q(x)$
 is true.



Domain: All students of xyz university.

$P(x)$: x has studied discrete mathematics.

$Q(x)$: x has scored more than 60% marks in exam.

$\exists x (P(x) \vee Q(x))$: Some students of xyz university has studied discrete mathematics or has scored more than 60% marks.

$\exists x P(x) \vee \exists x Q(x)$:
Some students of xyz university has studied discrete mathematics or some students of xyz university has scored more than 60% marks.

name	discrete maths	>60%
	✓	x
joe	x	x

Negation of Quantified statement

Rule

statement	Negation
1 $\forall x P(x)$ [all true]	1 $\exists x [\neg P(x)]$ [at least one false]
2 $\exists x [\neg P(x)]$ [at least one false]	2 $\forall x P(x)$ [all true]
3 $\forall x [\neg P(x)]$ [All false]	3 $\exists x P(x)$ [at least one true]
4 $\exists x P(x)$ [at least one true]	4 $\forall x [\neg P(x)]$ [all false]

Negate the statements **IMPORTANT-NOTES**

1 All integers are greater than 8.

sol $p(x)$: x is an integer $Q(x)$: x is greater than 8.

$$\therefore \forall x (p(x) \rightarrow Q(x)) \equiv \forall x (\neg p(x) \vee Q(x)).$$

Negation

$$\exists x \neg (\neg p(x) \vee Q(x))$$

$$\equiv \exists x (p(x) \wedge \neg Q(x)).$$

2 Not all men are genius.

sol $M(x)$: x is a man $G(x)$: x is genius

$$\neg [\forall x (M(x) \rightarrow G(x))] \equiv \forall x \neg (M(x) \rightarrow G(x))$$

Negation

$$\neg [\neg (\forall x (M(x) \rightarrow G(x)))]$$

$$\equiv \exists x (M(x) \rightarrow G(x))$$

3 $(\forall x p(x)) \vee (\exists y p(y))$

Negation

$$\neg [\forall x p(x) \vee \exists y p(y)]$$

$$\equiv \exists x (\neg p(x)) \wedge \forall y (\neg p(y)).$$

4 $\forall x \in \mathbb{R}, x > 5 \rightarrow x^3 > 125$

Negation

$$\exists x \in \mathbb{R}, x > 5 \text{ and } x^3 \leq 125.$$



De-MORGAN'S Law for Quantifiers

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$p \rightarrow$ propositional function . $D \rightarrow$ Domain of discourse.

$$a) \neg (\forall x p(x)) \equiv \exists x (\neg p(x)).$$

$$b) \neg (\exists x p(x)) \equiv \forall x (\neg p(x))$$

Proof a)

$$\neg (\forall x p(x)) \equiv \exists x (\neg p(x))$$

Let assume

$\neg (\forall x p(x))$ is true

" $\forall x p(x)$ is false.

There exists x such that $p(x)$ is f.

Therefore, there exists x such that $\neg p(x)$ is T.

$\exists x (\neg p(x))$ is True.

Now,

Let's assume $\neg (\forall x p(x))$ is false.

" $\forall x p(x)$ is True.

~~is it is true false~~

is for every x , $p(x)$ is true.

for every x , $\neg p(x)$ is false.

$\therefore \exists x (\neg p(x))$ is false.

we can show that

$\forall x p(x)$ is

True, $p(x)$ is True for every x .

False, $p(x)$ is false for at least one x .

$\exists x p(x)$ is

True, $p(x)$ is True for at least one x .

False, $p(x)$ is false for every x .

IMPORTANT NOTES

let assume

$\neg (\exists x p(x))$ is true.

$\therefore \exists x p(x)$ is false.

$\therefore$ ~~there~~ $p(x)$ is false for every x .

$\therefore$ ~~there exists~~ $\forall x (\neg p(x))$ is true.

Now,

let assume

$\neg (\exists x p(x))$ is false

$\therefore \exists x p(x)$ is true

$\therefore$ there exists x such that $p(x)$ is true

$\therefore$ for all x , $\neg p(x)$ is false

$\therefore \forall x (\neg p(x))$ is false

Converting English Sentences in Logical Expressions.

ex: "You are not allowed to watch adult movies if your age is less than 18 years or you have no age proof."

Step 1: Find the connectives which are connecting the two propositions together

Step 2: Rename the propositions

Let $q =$ "You are allowed to watch adult movies". **IMPORTANT-NOTE**

$r =$ "Your age is less than 18 years".

$s =$ "You have age proof".

$$\therefore (r \vee \neg s) \rightarrow \neg q.$$

Problem

Are these system specification consistent?

"If system is in multiuser state if and only if it is operating normally".

"If the system is operating normally, then the kernel is functioning."

"The kernel is not functioning or the system is in interrupt mode".

"If the system is not in multiuser state, then it is in interrupt mode."

"The system is not in interrupt mode".

Sol

Let

$p =$ "The system is in multiuser state"

$q =$ "The system is operating normally"

$r =$ "The kernel is functioning"

$s =$ "The system is in interrupt mode"

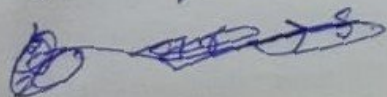
1 $p \leftrightarrow q \rightarrow \text{True}$

2 $\neg r \vee s \rightarrow \text{True}$

3 $q \rightarrow r \rightarrow \text{True}$

4 $\neg p \rightarrow s \rightarrow \text{True}$

5 $\neg s \rightarrow \text{True}$



Consistent

means assigning truth values to propositional variables in such a way that finally we would be able to make all specifications "true".

IMPORTANT-NOTES

Tip → Always start from the statement which involves least no. of prepositions.

We cannot make P ~~Q~~ false

a system specification is not consistent.

Q "Some students in the XYZ University has studied discrete mathematics"

Sol Let us consider the domain consists of all students of XYZ University.

Domain: All students of XYZ University.

$P(x)$: x has studied discrete mathematics

$$\exists x P(x)$$

Let's change the Domain

Domain: All people on the planet earth.

$S(x)$: x is a student of XYZ University

$P(x)$: x has studied discrete mathematics.

$$\exists x (S(x) \wedge P(x))$$

Types

Problem

Consider These statements, The first two are called premises. and the third is called conclusion. The entire set is called an argument.

"All lions are fierce"

"Some lions do not drink coffee"

"Some fierce creatures do not drink coffee"

IMPORTANT-NOTE

Let $P(x)$, $Q(x)$ and $R(x)$ be the statements "x is a lion", "x is a fierce", and "x drinks coffee" respectively. Assuming that the domain consists of all creatures, express the statements in the argument using predicates $P(x)$, $Q(x)$ and $R(x)$.

eg)

Domain: All creatures.

$P(x)$: x is a lion.

$Q(x)$: x is a fierce.

$R(x)$: x drinks coffee.

$$\forall x (P(x) \rightarrow Q(x))$$

$$\exists x (P(x) \wedge \neg R(x))$$

$$\exists x (Q(x) \wedge \neg R(x))$$

NESTED Quantifiers

Two quantifiers are said to be nested if one is within the scope of the other.

eg $\forall x \exists y Q(x, y)$

$\exists$ is within the scope of $\forall$

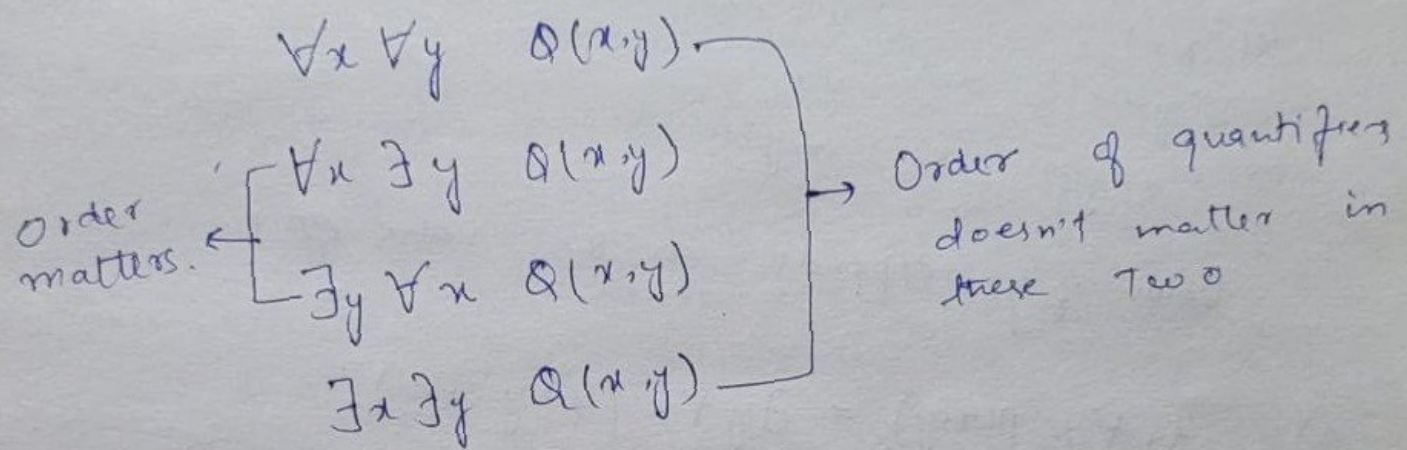
NOTE: Any thing within a scope of the quantifier can be thought of as a propositional function.

IMPORTANT-NOTES

Let $\exists x Q(x,y) = P(x)$

$\therefore \forall x \exists y Q(x,y) = \forall x P(x).$

Different combinations of Nested Quantifiers



i.e. $\forall x \forall y Q(x,y) \equiv \forall y \forall x Q(x,y).$

$\exists x \exists y Q(x,y) \equiv \exists y \exists x Q(x,y).$

Ex 1

Let x and y be the real numbers and $P(x,y)$ denotes " $x + y = 0$ ". Find the values of.

- a) $\forall x \forall y P(x,y)$
- b) $\forall x \exists y P(x,y)$
- c) $\exists y \forall x P(x,y)$
- d) $\exists x \exists y P(x,y)$

Sol: a) Domain: All real numbers

$\forall x \forall y P(x,y) \equiv \forall x \forall y (x + y = 0)$

for all real numbers x and y , $x + y = 0$.

It is clear that for all real numbers x & y , $x + y \neq 0$.
 $\therefore$ Truth value is **F**.



IMPORTANT-NOTES

b) $\forall x \exists y P(x, y) \equiv \forall x \exists y (x+y=0)$
 for every real number x , "there exists a real number y such that $x+y=0$ ".

It is true.

let $x=1$	$x=-1$
$\therefore y=-1$	$y=1$
$\therefore x+y=0$	$\therefore x+y=0$

Value of y depends on the value of x .

c) $\exists y \forall x P(x, y) \equiv \exists y \forall x (x+y=0)$

"There exists some real number y such that for every real number x , $x+y=0$ ".

It is not true.

because, it is asking us to find a real number y for which $P(x, y)$ becomes true by plugging in every real number x .

d) $\exists x \exists y P(x, y) \equiv \exists x \exists y (x+y=0)$

"There exist some real number x and y such that $x+y=0$ ".

It is true.

as there are some combinations of real numbers x & y exist for which $x+y=0$.

IMPORTANT NOTES

Q7 Let $S(x, y, z)$ be the statement " $x + y = z$ ".
What are the truth values of the statements
 $\forall x \forall y \exists z S(x, y, z)$ and $\exists z \forall x \forall y S(x, y, z)$,
where the domain of all variables consists
of all real numbers?

Sol
(i) $\forall x \forall y \exists z S(x, y, z) \equiv (z = x + y) \forall x \forall y$

"for all real numbers x and for all
real numbers y , there exists number z
such that $x + y = z$ " is true.

Let $x = 1$ & $y = -1$
then $z = 0$.

i. It is clear that for every combination of
real numbers x & y , there is always a real
number z such that $x + y = z$.

$\therefore \forall x \forall y \exists z (x + y = z)$ is true.

(ii) $\exists z \forall x \forall y S(x, y, z) \equiv (z = x + y) \exists z \forall x \forall y$

"there exist some real number z such that
for all real number x and y , $x + y = z$ "

Let $z = 2$

for $x = 1$ & $y = 1$, $x + y = z$ is true.

But for $x = 2$ & $y = 3$, $x + y = z$ is not true.

It is clear that there is no real number z
such that $x + y = z$ becomes true for all real
numbers x and y .

$\therefore \exists z \forall x \forall y (x + y = z)$ is false.



Negation of nested Quantifiers

IMPORTANT-NOTE

Understanding how to negate the nested quantifiers is crucial.

Rule for Negating Nested Quantifiers

1. Negate the outermost quantifier.

• If the outermost quantifier is $\forall$, change it to $\exists$, and vice versa.

2. Negate the inner quantifier

• Repeat step 1 for the next innermost quantifier.

3. Negate the predicate.

• Negate the proposition or predicate that follows the innermost quantifier.

Ex

"For every person, there exists a book that they have read"

In logical notation:

$$\forall x \exists y (\text{Reads}(x, y))$$

To negate this statement:

1. Negate the outer quantifier ($\forall$): $\exists x$

2. Negate the inner quantifier ($\exists$): $\forall y$

3. Negate the predicate ($\text{Reads}(x, y)$): $\neg \text{Reads}(x, y)$

$\therefore$ The negated statement becomes:

$$\exists x \forall y \neg \text{Read}(x, y)$$

"There exists a person such that for every book, they have not read it".

Rules of Inference.

IMPORTANT-NOTES

These are the templates for constructing valid arguments.

Inference $\rightarrow$ means deriving conclusions from ~~the~~ evidences.

Types

1. Modus Ponens Rule

$$\begin{array}{c} p \rightarrow q \\ p \\ \hline \therefore q \end{array} \quad \text{OR} \quad [(p \rightarrow q) \wedge p] \rightarrow q.$$

2. Modus Tollens Rule

$$\begin{array}{c} p \rightarrow q \\ \sim q \\ \hline \sim p \end{array}$$

3. Addition

$$a) \frac{p}{p \vee q}$$

$$b) \frac{q}{p \vee q}$$

4. Simplification

$$a) \frac{p \wedge q}{p}$$

$$b) \frac{p \wedge q}{q}$$



5 Conjunction

$$\begin{array}{c} p \\ q \\ \hline p \wedge q \end{array}$$

6 Hypothetical
syllogism

$$\begin{array}{c} p \rightarrow q \\ q \rightarrow r \\ \hline p \rightarrow r \end{array}$$

7 Disjunctive
syllogism

$$\begin{array}{c} p \vee q \\ \neg p \\ \hline q \end{array}$$

8 Constructive dilemma.

$$\begin{array}{c} (p \rightarrow q) \wedge (r \rightarrow s) \\ p \vee r \\ \hline q \vee s \end{array}$$

9 Destructive
dilemma

$$\begin{array}{c} (p \rightarrow q) \wedge (r \rightarrow s) \\ \neg q \vee \neg s \\ \hline \neg p \vee \neg r \end{array}$$

IMPORTANT NOTES

Q1 If Ram works hard, he will get a job.

Ram works hard. Therefore, he will get a job.

Check validity.

Sol

p : Ram works hard.

q : He will get a job.

premises: $p \rightarrow q$, p conclusion: q .

1. $p \rightarrow q$ premise.

2. p premise

$p \rightarrow q$	$\neg$
p	
q	
$\rightarrow$ Modus ponens.	

3. q Modus ponens using 1 and 2.

$\therefore$ the argument is valid.

Q2 If a man is a bachelor, he is unhappy, If a man is unhappy, he dies young. Therefore, bachelor dies young. Check validity.

Sol

p : A man is a bachelor.

q : A man is unhappy.

r : A man dies young.

premises: $p \rightarrow q$, $q \rightarrow r$.

conclusion $p \rightarrow r$.

1. $p \rightarrow q$ $\rightarrow$ premise

2. $q \rightarrow r$ $\rightarrow$ premise

$p \rightarrow q$	hypothetical
$q \rightarrow r$	syllogism
$p \rightarrow r$	

3. $p \rightarrow r$ Hypothetical syllogism using (1) and (2)

$\therefore$ argument is valid

IMPORTANT-NOTE

Q If this number is divisible by 6, then it is divisible by 3. This number is not divisible by 3. Therefore, this number is not divisible by 6.

p : This number is divisible by 6.

q : ~~Q~~ This number is divisible by 3.

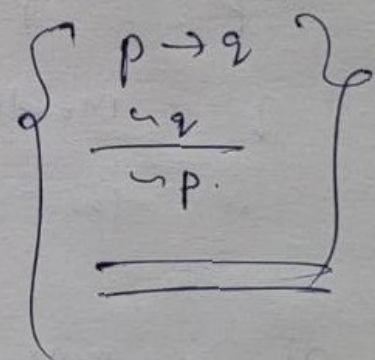
~~$\neg q$: This number is not divisible~~

Premises $\Rightarrow$

$p \rightarrow q$ $\neg q$

Conclusion

$\neg p$.



Premise

1. $p \rightarrow q$

2. $\neg q$.

3. $\neg p$. { ~~Q~~ modus tollens rule using ① & ② }

The argument is valid.

Q Either Ram is not guilty or Shyam is telling the truth. Shyam is not telling the truth. $\therefore$ Ram is not guilty.

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p : Ram is not guilty

q : Shyam is telling truth.

Premises

$p \vee q$

$\neg q$.

Conclusion

p .

IMPORTANT NOTES

$$1. \quad p \vee q.$$

$$2. \quad \neg q.$$

3. p { Disjunctive Syllogism using ① & ② }
is argument is valid.

Q Show that t is a valid conclusion from the premises $p \rightarrow q$, $q \rightarrow r$, $r \rightarrow s$, $\neg s$ and $p \vee t$.

Sol 1. $p \rightarrow q$ premise

2. $q \rightarrow r$ premise

3. $p \rightarrow r$ Hypothetical Syllogism using ① & ②.

4. $r \rightarrow s$ premise

5. $p \rightarrow s$ Hypothetical Syllogism using ③ & ④.

6. $\neg s$ premise.

7. $\neg p$ Modus tollens using ⑤ & ⑥.

8. $p \vee t$ premise

9. t Disjunctive Syllogism using ⑦ & ⑧.

• Hence, t is a valid conclusion.

IMPORTANT-NOTE

Q. Show that S is a valid conclusion from the premises $p \rightarrow q$, $p \rightarrow r$, $\neg(q \wedge r)$ and $S \vee p$.

- Sol.
1. $p \rightarrow q$ premise
 2. $p \rightarrow r$ premise
 3. Taking $\wedge$ b/w ① & ②
 4. $(p \rightarrow q) \wedge (p \rightarrow r)$ using ① & ②.
 5. $\neg(q \wedge r)$ premise.
 6. $\neg p \vee \neg p$ (De Morgan's Law ~~using~~ using ④)
 7. $\neg p \vee \neg p$ of Destructive Dilemma using ③ & ⑤
 8. $\neg p$ Idempotent law.
 9. $S \vee p$ premise.
 10. S of Destructive syllogism using ⑧ & ⑨

Hence, S is a valid argument.